
Current Position and Contact Information

Professor, Department of Mathematical Sciences, School of Science, Tokai University
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Research Interests

Algebraic geometry; Hilbert schemes; deformation theory of embedded curves; obstructions to deformations; space curves; K3 surfaces; del Pezzo surfaces; Fano varieties; non-reduced components of Hilbert schemes.

Education and Degrees

Apr. 1993–Mar. 1997	Undergraduate studies in Mathematics, Nagoya University, Japan.
Apr. 1997–Mar. 1999	Master's program, Graduate School of Mathematics, Nagoya University, Japan. Master of Mathematical Science. Master's thesis: Chow stability of space curves. Advisor: Prof. Shigeru Mukai.
Apr. 1999–Mar. 2001	Doctoral program, Graduate School of Mathematics, Nagoya University, Japan.
Apr. 2001–Mar. 2005	Doctoral program, Graduate School of Science, Kyoto University, Japan. Ph.D. in Science, March 2005. Doctoral thesis: <i>Obstructions to deforming space curves and non-reduced components of the Hilbert scheme</i> . Advisor: Prof. Shigeru Mukai.

Academic Appointments

Apr. 2005–Mar. 2008	Research Fellow (COE) and Part-time Teaching Associate, Research Institute for Mathematical Sciences (RIMS), Kyoto University, Japan.
Apr. 2008–Sep. 2008	Visiting Researcher, Department of Mathematics, University of Oslo, Norway (Research Council of Norway / JSPS Fellowship 2008).
Oct. 2008–Mar. 2011	Assistant Professor, School of Information Environment, Tokyo Denki University, Japan.
Apr. 2010–Mar. 2011	Part-time Lecturer, School of Fundamental Science and Engineering, Waseda University, Japan.
Apr. 2011–Mar. 2014	Lecturer, Department of Mathematical Sciences, Tokai University, Japan.
Apr. 2014–Mar. 2021	Associate Professor, Department of Mathematical Sciences, Tokai University, Japan.
Apr. 2019–Sep. 2019	Visiting Researcher, Department of Mathematics, University of Oslo, Norway.
Apr. 2021–present	Professor, Department of Mathematical Sciences, Tokai University, Japan.

Peer-Reviewed Publications

1. H. Nasu, “Obstructions to deforming space curves and non-reduced components of the Hilbert scheme”, *Publ. Res. Inst. Math. Sci.* **42** (2006), 117–141.
2. H. Nasu, “The Hilbert scheme of space curves of degree d and genus $3d - 18$ ”, *Communications*

in *Algebra* **36** (2008), no. 11, 4163–4185.

3. H. Nasu, “Deformations of degenerate curves on a Segre 3-fold”, in *Higher Dimensional Algebraic Varieties and Vector Bundles*, RIMS Kôkyûroku Bessatsu **B9**, RIMS, Kyoto, 2008, 163–174.
4. S. Mukai and H. Nasu, “Obstructions to deforming curves on a 3-fold, I: A generalization of Mumford’s example and an application to Hom schemes”, *Journal of Algebraic Geometry* **18** (2009), 691–709.
5. H. Nasu, “Obstructions to deforming curves on a 3-fold, II: Deformations of degenerate curves on a del Pezzo 3-fold”, *Annales de l’Institut Fourier* **60** (2010), no. 4, 1289–1316.
6. H. Nasu, “Obstructions to deforming curves on a 3-fold, III: Deformations of curves lying on a K3 surface”, *International Journal of Mathematics* **28** (2017), no. 13, 1750099, 30 pp.
7. H. Nasu, “Obstructions to deforming curves on a prime Fano 3-fold”, *Mathematische Nachrichten* **292** (2019), no. 8, 1777–1790.
8. H. Nasu, “Corrigendum to ‘Obstructions to deforming curves on a 3-fold, III: Deformations of curves lying on a K3 surface’ ”, *International Journal of Mathematics* **31** (2020), no. 12, 2092001, 6 pp.
9. H. Nasu, “Obstructions to deforming curves on an Enriques-Fano 3-fold”, *Journal of Pure and Applied Algebra* **225** (2021), no. 9, 106677, 15 pp.
10. H. Nasu, “Obstructions to deforming space curves lying on a smooth cubic surface”, *Manuscripta Mathematica* **172** (2023), nos. 1–2, 31–55.
11. H. Nasu, “Obstructions to deforming space curves lying on a del Pezzo surface”, *Manuscripta Mathematica* **177** (2026), Article 33, 30 pp.

Proceedings Articles and Other Scholarly Publications

1. H. Nasu, “Obstructions to deforming space curves and non-reduced components of the Hilbert scheme” (in Japanese), proceedings of the Kinosaki Algebraic Geometry Symposium, 2004, 41–49; proceedings of *Algebraic and Topological Geometry*, 2004, 61–70; and proceedings of the 2nd Symposium on Algebraic Curves, 2004, 53–61.
2. S. Mukai and H. Nasu, “A new example of a generically non-reduced component of the Hom scheme” (in Japanese), proceedings of *Geometry of Projective Varieties and Related Topics 2005*, 2005, 45–53.
3. S. Mukai and H. Nasu, “The Hilbert scheme of canonical curves on del Pezzo 3-folds and its application to Hom scheme”, proceedings of the 3rd Symposium on Algebraic Curves, 2005, 102–113.
4. H. Nasu, “Deformations of degenerate curves on a del Pezzo 3-fold” (in Japanese), proceedings of *Geometry of Projective Varieties and Related Topics 2006*, 2006, 69–80.
5. H. Nasu, “Obstructions to deforming space curves and a remark to a conjecture of Kleppe”, proceedings of the 6th Symposium on Algebraic Curves, 2008, 79–86.
6. H. Nasu, “Obstructions to deforming space curves and a remark to a conjecture of Kleppe” (in Japanese), proceedings of *Geometry of Projective Varieties and Related Topics 2009*, 2009, 1–10.
7. H. Nasu, “Obstructions to deforming degenerate curves on a scroll” (in Japanese), proceedings of the Miyakonoseihoku Algebraic Geometry Symposium, 2010, 105–114.

8. H. Nasu, “Obstructions to deforming space curves lying on a smooth quartic surface”, proceedings of the 11th Symposium on Algebraic Curves, 2014, 105–117.
9. H. Nasu, “Obstructions to deforming space curves lying on an elliptic quartic surface” (in Japanese), *RIMS Kôkyûroku*, No. 1897 (2014), 128–141.
10. H. Nasu, “Obstructions to deforming curves lying on a K3 surface in a Fano 3-fold” (in Japanese), proceedings of *Geometry of Projective Varieties and Related Topics 2015*, 2015, 31–44.
11. H. Nasu, “Obstructions to deforming curves lying on a K3 surface in a Fano 3-fold”, proceedings of the Miyakonoseihoku Algebraic Geometry Symposium, 2016, 25–37.
12. H. Nasu, “Obstructions to deforming curves on an Enriques-Fano 3-fold”, proceedings of the Algebraic Geometry Symposium, 2019, 85–94.
13. H. Nasu, “Primary obstructions to deforming embedded curves and an application to the Hilbert scheme of space curves”, proceedings of the 20th Symposium on Algebraic Curves, 2022.
14. H. Nasu, “Obstructions to deforming space curves lying on a smooth del Pezzo surface” (in Japanese), proceedings of *Algebraic Geometry and Related Topics*, 2024.

Research Grants and Competitive Funding

- **2024–2028**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Scientific Research (C), No. 24K06677.
Explicit study of Hilbert schemes of higher-dimensional algebraic varieties using intermediate surfaces and morphisms.
- **2020–2023**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Scientific Research (C), No. 20K03541.
Study on Hilbert schemes from the viewpoint of morphisms of algebraic varieties.
- **2017–2020**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Scientific Research (C), No. 17K05210.
Obstructions to deforming curves on algebraic varieties and a study on their Hilbert schemes.
- **2013–2015**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Scientific Research (C), No. 25400048.
Deformations of curves on a uniruled variety and their obstructions.
- **2013–2018**, Collaborating Researcher, JSPS KAKENHI, Grant-in-Aid for Scientific Research (S), No. 25220701 (Principal Investigator: Shigeru Mukai, Kyoto University).
Moduli spaces of algebraic varieties and self-morphisms.
- **2011–2012**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Young Scientists (B), No. 23740032.
Deformations of curves on a higher-dimensional algebraic variety and their obstructions.
- **2009–2010**, Principal Investigator, JSPS KAKENHI, Grant-in-Aid for Young Scientists (B), No. 21740029.
Deformations of curves on a higher-dimensional algebraic variety and their obstructions.

Organization of Conferences and Workshops

- **Mar. 19, 2026**, Co-organizer, *Algebra and Discrete Geometry*, Tokai University, Shibuya Campus, Tokyo. Co-organizers: Shingo Taki (Tokai University) and Toru Tsukioka (Nihon University).

- **Feb. 20–21, 2026**, Organizer, *Tokai Workshop in Algebraic Geometry 2026*, Tokai University, Shonan Campus, Kanagawa. Research workshop in algebraic geometry with invited speakers from Japan and Korea; followed by a research meeting in Hakone, Feb. 22–23.
- **Oct. 20–23, 2020**, Co-organizer, *Kinosaki Algebraic Geometry Symposium 2020*, online (Zoom). Co-organizers: Atsushi Ito (Nagoya University) and Hiroshi Sato (Fukuoka University). Supported in part by JSPS KAKENHI No. 20K03541 (Nasu).
- **Mar. 11–15, 2019**, Member of the Organizing Committee, *Algebraic Geometry and Moduli Theory — Conference in Honor of Shigeru Mukai’s Retirement*, Research Institute for Mathematical Sciences (RIMS), Kyoto University. Co-organizers: Shigeyuki Kondo (Nagoya University), Noboru Nakayama (Kyoto University), and Hisanori Ohashi (Tokyo University of Science). Co-hosted by RIMS, Kyoto University.
- **Aug. 29–31, 2013**, Co-organizer, *Workshop in Algebraic Geometry in Sapporo*, Hokkaido University, Sapporo. Co-organizer: Hokuto Uehara (Tokyo Metropolitan University).

Selected Invited Talks and Lecture Series (Recent)

- **Sep. 17–18, 2026**, *Obstructions to deforming curves lying on a complete intersection of quadrics*, three 75-minute lectures and one 75-minute research talk, KU Lectures in Algebraic Geometry 2026: *Deformation Theory of Embedded Curves and Its Applications*, Korea University, Seoul, Korea.
- **Mar. 4, 2026**, *Deformations of space curves lying on a complete intersection K3 surface of genus 5*, 24th Affine Algebraic Geometry Meeting, Hirosaki University, Hirosaki, Japan.
- **Jan. 26, 2026**, *Obstructions to deforming space curves lying on a complete intersection of quadrics*, Birational and Affine Geometry, Songdo, Incheon, Korea.
- **Jun. 12, 2025**, *Deformations of space curves lying on a del Pezzo surface*, 2025 Algebraic Geometry Conference in Jeonju Hanok Village, Jeonju, Korea.
- **Jun. 11, 2025**, *Obstructions to deforming curves lying on a K3 surface*, Department Seminar/Colloquium, KAIST, Daejeon, Korea.
- **Mar. 5, 2025**, *Stably degenerate and obstructed curves on a del Pezzo surface*, 23rd Affine Algebraic Geometry Meeting, Niigata University, Niigata, Japan.
- **Feb. 5, 2025**, *Obstructions to deforming curves on Fano threefolds and fourfolds*, Workshop on Fano Varieties and Birational Geometry, Nagoya University, Nagoya, Japan.

Professional Service

Apr. 2026–present Vice President, *Tokyo Journal of Mathematics*.
 Apr. 2025–present Board Member, *Tokyo Journal of Mathematics*.
 Member, Mathematical Society of Japan.

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